

Guide For Engineers To Create INNOVATIVE, MANUFACTURABLE PCBs

Designing a printed circuit board that works on paper is one thing, making it production-ready is the real test. How it performs across tools, teams, and real-world conditions is what turns a good design into a great product. Part 1 of this article was published in December 2024 issue of the magazine.



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Design is only the beginning when it comes to printed circuit boards (PCBs). Beyond placing components and routing traces lies the challenge of turning concepts into manufacturable, high-performance products. Success demands foresight, collaboration across functions, and a deep understanding of production realities. Building on the fundamentals from Part 1, this piece explores how to align design with fabrication limits, simulate for reliability, and manage mechanical constraints—arming engineers with tools to bring robust, production-ready PCBs to life.

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Key areas covered here include computer-aided manufacturing (CAM) and its impact on design-to-production alignment; the role of ECAD-MCAD collaboration in meeting mechanical constraints; the value of simulation and iterative design refinement for performance and reliability; and practical strategies to manage variations in the PCB manufacturing process.

1. Computer aided manufacturing (CAM). Front-end CAM tooling departments play a critical role in design-for-manufacture. One of their most frequent tasks is editing electrical and non-electrical layers—changes that significantly impact manufacturing yield and product reliability. These are often not considered during the design phase.

Additional CAM responsibilities include solder mask adjustments to ensure proper

clearances around solderable features, preventing issues such as solder bridging or insufficient pad exposure. CAM engineers at the PCB fabrication house may also recommend design modifications before production. These recommendations are based on vendor-specific capabilities, which vary from one manufacturer to another.

Designers should therefore understand the fabrication capabilities of their chosen PCB house during the design phase to avoid time-consuming iterations. A lack of detailed fabrication instructions can lead to CAM-stage challenges that are otherwise preventable.

To avoid unexpected CAM modifications, designers should consider the following based on fabrication capabilities:

- Increase trace-to-trace spacing to prevent shorts
- Maintain adequate solder mask dam clearance to meet process standards
- Adjust pad shapes (e.g. teardrops) to maintain annular ring integrity
- Apply etch compensation for accurate trace dimensions
- Remove copper from non-plated holes, add copper plating to balance electroplating, and modify stack-ups for impedance control

2. Importance of ECAD and MCAD collaboration. ECAD and MCAD collaboration becomes essential when PCBs face mechanical constraints. This is particularly relevant for applications where multiple PCB assemblies must integrate with enclosure designs or interconnect